Daily Integral - Hard

$$\int_{-\pi}^{\pi} \cos(\sin(x))e^{\cos(x)}\,dx$$

This problem was generously submitted by Leonz on https://dailyintegral.com.

Solution

Since by Euler's formula, $e^{ix} = \cos(x) + i\sin(x)$, we can find that $e^{e^{ix}} = e^{\cos(x)+i\sin(x)} = e^{\cos(x)}e^{i\sin(x)}$. Using Euler's formula again, we get that $e^{e^{ix}} = e^{\cos(x)}(\cos(\sin(x) + i\sin(\sin(x))$. Hence, the integrand is simply the real part of the expression, and we can make the substitution

$$\begin{aligned}&\int_{-\pi}^{\pi} \Re\left(e^{e^{ix}}\right)\,dx\\&= \Re\left(\int_{-\pi}^{\pi} e^{e^{ix}}\,dx\right).\end{aligned}$$

To evaluate this integral, we use Feynman's trick to introduce a parameter:

$$I(t) = \Re\left(\int_{-\pi}^{\pi} e^{te^{ix}}\,dx\right).$$

Note that the original integral is given when $t = 1$. Now, we can differentiate with respect to t:

$$\begin{aligned}I'(t) &= \Re\left(\int_{-\pi}^{\pi} \frac{\partial}{\partial t} e^{te^{ix}}\,dx\right)\\&= \Re\left(\int_{-\pi}^{\pi} e^{ix}e^{te^{ix}}\,dx\right).\end{aligned}$$

Now, take $u = e^{ix}$. Then, $1/i\,du = e^{ix}\,dx$, and $u \in [-1, -1]$:

$$\begin{aligned}I'(t) &= \Re\left(\frac{1}{i}\int_{-1}^{-1} e^{tu}\,du\right)\\&= 0.\end{aligned}$$

Now we can find that

$$\begin{aligned}I(t) &= \int I'(t)\,dt\\&\int 0\,dt\\&= C.\end{aligned}$$

Since $I(t)$ equals some constant C for all t, we can see that $I(0) = I(1)$, which will give us the value of the original integral:

$$\begin{aligned}I(0) &= \Re\left(\int_{-\pi}^{\pi} e^{0e^{ix}}\,dx\right)\\&= \Re\left(\int_{-\pi}^{\pi} dx\right)\\&= 2\pi.\end{aligned}$$